

PROJECT REQUIREMENTS DOCUMENT

Project title : Android App Development using GenAI - Kreeda-Ankana (Sports)

Name : Namith K P
USN : 1VA22IS058
College : SAI VIDYA INSTITUTE OF TECHNOLOGY
Email ID : namithkp.22is@saividya.ac.in

1. Introduction

1.1 Purpose

The purpose of this project is to design and develop a mobile application that enables sports teams in villages and local communities to book ground slots and organize friendly matches. The application aims to bring structure and accessibility to informal sports culture by providing a digital platform for ground management.

1.2 Scope

The Kreeda-Ankana app will:

- Allow teams to register and book ground time slots
- Enable teams to post and respond to match challenges
- Provide a real-time ground calendar visible to all users
- Display a score wall for recording village match results

1.3 Definitions

- **Ground:** A sports venue (Volleyball/Cricket) available for booking
- **Slot:** A reserved time window at the ground (e.g., 4 PM – 6 PM)
- **Challenge:** A public invitation posted by a team seeking an opponent
- **Score Wall:** A community board to post and view match results

2. Problem Statement

In many villages and rural areas:

- Good sports grounds exist but are monopolized by the same group all day
- There is no system to "Book a Slot" or "Find an Opponent"
- Small teams have no way to organize a Friendly Match with the next village
- There is no centralized platform for ground scheduling or match coordination

This project solves this gap using a mobile-based sports ground management solution.

3. Objectives

Primary Objectives

- Build a simple and user-friendly Android sports management app
- Enable ground slot booking with conflict prevention
- Support community match challenges and real-time updates

Secondary Objectives

- Promote sports culture and community engagement
- Introduce AI-based match suggestions and team insights
- Provide scalable architecture for future sport types

4. Stakeholders

- **End Users:** Village sports teams and ground managers
- **Developers:** Android developers
- **Evaluators:** Faculty / internship reviewers
- **Community:** Local users benefiting from organized sports

5. Functional Requirements

5.1 Team Registration

Description: Teams can create their profiles.

Inputs:

- Team name
- Captain's phone number
- Sport type (Volleyball / Cricket / Other)
- Village / Location
- Preferred playing time

Processing:

- Validate inputs
- Store data in Firebase
- Output: Confirmation message: 'Team Registered Successfully'

5.2 Ground Calendar (Slot View)

Description: Users can view the ground schedule for the day/week.

Features:

- Grid-based calendar showing all booked slots
- Color coding: booked vs available slots

Processing:

- Query Firebase for current bookings
- Render a time-grid with slot status
- Output: Visual calendar of the ground

5.3 Book a Slot

Description: Teams can reserve a time slot at the ground.

Inputs:

- Team name
- Date
- Start time and End time

Processing:

- Check for slot conflicts in Firebase
- If available, confirm booking; if occupied, show conflict error

- Output: Booking confirmation or conflict notification

5.4 Challenge Board

Description: Teams can post and reply to match challenges.

Inputs:

- Challenge message (e.g., "Our team is ready for Volleyball this Sunday")
- Sport type
- Proposed date and time

Processing:

- Post challenge to Firebase real-time database
- Other teams can reply / accept the challenge
- Output: Live challenge thread visible to all users

5.5 Score Wall

Description: Display and record results of village matches.

Inputs:

- Winning team and losing team
- Score
- Date of match
- Output: Stored and displayed on the Score Wall

6. GenAI Integration

Feature 1: Smart Match Suggestions

Input: Team profile and preferences; Output: AI-suggested opponents and suitable time slots

Feature 2: Auto Challenge Message Generator

Input: Sport type, date, and time; Output: AI-generated challenge post for the Challenge Board

7. Non-Functional Requirements

7.1 Performance

Fast loading (< 2 seconds response time for slot availability checks)

7.2 Usability

Bold and sporty UI; simple navigation suitable for rural users

7.3 Reliability

App should not crash; booking conflicts must be handled gracefully

7.4 Security

Data stored securely in Firebase; booking actions require team login

7.5 Scalability

Support multiple grounds and increasing number of teams over time

8. System Architecture

Architecture Type: Client-Server Architecture

Frontend

Android App developed using Kotlin/Java

Backend

Firebase Firestore (NoSQL database) for bookings, challenges, and scores; Firebase Realtime Database for the live Challenge Board; Firebase Authentication (optional)

Data Flow

- User inputs booking or challenge data
- App sends request to Firebase
- Firebase checks conflicts / stores data and returns response
- App displays updated calendar or challenge board

9. Database Design

Collection: teams

Field	Type	Description
id	String	Unique team ID
teamName	String	Name of the team
captain	String	Captain contact
sport	String	Sport type
location	String	Village / Area
preferredTime	String	Preferred slot

Collection: bookings

Field	Type	Description
bookingId	String	Unique booking ID
teamId	String	Reference to team
date	String	Date of booking
startTime	String	Slot start time
endTime	String	Slot end time

Collection: challenges

Field	Type
challengeId	String
teamId	String
message	String
sport	String
proposedDate	String

status	String (open / accepted)
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10. UI/UX Design Requirements

- Bold and sporty design with vibrant colors
- Grid-based calendar for slot visualization
- High readability and simple navigation

Screens:

- Home Screen
- Team Registration
- Ground Calendar (Slot View)
- Book a Slot Screen
- Challenge Board Screen
- Score Wall Screen

11. User Flow

- User launches the Kreedaa-Ankana app
- Chooses: Register Team or View Ground Calendar
- Registered team selects a free slot and books it
- Team posts a challenge on the Challenge Board
- Another team views and replies/accepts the challenge
- Match results are posted on the Score Wall

12. Testing Strategy

Types of Testing:

- Unit Testing
- Integration Testing
- UI Testing

Test Cases:

- Register a team
- Book a slot and verify conflict prevention
- Post and reply to a challenge
- Add a score wall entry
- Error handling for duplicate bookings

13. Deliverables

- APK file
- Source Code
- Documentation (PRD + Report)
- Presentation Slides
- Demo Video

14. Success Criteria

- The "Booking" feature must prevent two teams from picking the same time slot

- The "Challenge Board" must allow users to "Reply" to a challenge
- The UI must be bold and sporty
- Firebase real-time updates work correctly
- App functions without major bugs

15. Limitations

- Requires internet connection
- No real-time in-app chat (basic version)
- Limited AI capabilities in initial release
- Single ground per deployment in base version

16. Future Enhancements

- In-app team chat
- Live match score updates
- Push notifications for booking confirmations and challenge replies
- Advanced AI-based opponent and slot recommendations
- Multi-ground and multi-sport support
- Multi-language expansion for regional reach

17. Conclusion

The Kreedankana application is a practical solution to bring order and accessibility to village sports grounds. By enabling slot booking, real-time challenge posting, and score tracking, it transforms informal village play into an organized community sports culture. It leverages mobile technology and cloud services to promote sports, youth harmony, fitness, and community engagement in an efficient and scalable way.